

Investigation of Body-Part Authentication Performance Using On-Body Active Acoustic Sensing

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Abstract

Wearable devices can collect sensitive biometric and behavioral information, necessitating authentication. Common smartphone biometrics include fingerprint and facial recognition; their application to wearable devices is challenging due to varied attachment and limited input. This study investigates active acoustic sensing—a technique that analyzes responses from transmitted acoustic signals—as a method for user identification on wearable devices. We evaluated its performance across eight attachment sites: finger, wrist, neck, nose, earlobe, ear bone, teeth, and sole. Data were collected from seven participants over one week at three times of day (morning, midday, evening). Three evaluations were carried out to assess intra-day performance, generalization across days, and long-term stability. The results showed that the acquired biometric signals were unstable over time, leading to poor authentication performance at all sites, thus highlighting a significant challenge for wearable acoustic biometrics that motivates further research.

CCS Concepts

• Human-centered computing → Ubiquitous computing.

Keywords

Active Acoustic Sensing, Biometrics, Wearable

ACM Reference Format:

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1 Introduction

Wearable devices capture sensitive biometric and behavioral data, and reliable user authentication is vital, yet diverse attachment sites and limited interfaces hinder directly porting the biometric methods used on phones and laptops.

Active acoustic sensing is a technique in which acoustic signals are transmitted to a target, and the response, after propagating through the object, is analyzed to estimate its characteristics or

state. Researchers have recently begun applying this technology to the human body for user authentication purposes [4, 5]. However, because body tissue structures and sensor contact conditions vary by attachment site, the stability of authentication performance and comparative evaluations across different body locations have not been sufficiently examined. In this study, we apply active acoustic sensing to eight body sites and evaluate authentication performance by site and its variation over time of day. Data were collected from seven participants at eight sites (finger, wrist, neck, nose, earlobe, ear bone, teeth, sole) three times a day (morning, midday, evening) over seven days.

2 Related Works

This section introduces research on user identification using acoustics. Iwakiri et al.[5] acquired the response of sweep signals using a ring device equipped with a speaker and a microphone, reporting an EER of 0.027. Isobe et al.[4] mounted a speaker and a microphone on the nose pads of eyeglasses, and achieved an EER of 0.09 in an evaluation involving eleven participants. Arakawa et al.[2] proposed a user authentication method using reflections of sound transmitted into the ear canal and achieved an EER of 0.97. Konishi et al.[6] developed an authentication system that identifies users based on doorknob operation sounds, reporting an EER of 4.76% under a one-vs-rest setting. Chen et al.[3] proposed TouchPrint, which estimates the hand shape from multipath echoes during touch. Amesaka et al.[1] captured sound leaking from a hearable device using an external microphone and reported authentication accuracy ranging from 87.0% to 96.7%. Kubo et al.[7, 8] proposed an authentication system using the internal resonance characteristics of 3D-printed objects, achieving 92.2% accuracy. Unlike previous acoustic authentication studies, which predominantly evaluated their methods within a single day and did not fully address day-to-day performance variations, this study explicitly analyzes the impact of cross-day data variability on authentication accuracy.

3 Evaluation Method

To analyze authentication performance for each attachment site in detail, we use two methods: a distance-based authentication approach and a CNN+Transformer classification approach. This dual-method strategy provides both a straightforward baseline and an advanced learning model, allowing us to compare a simple distance metric against a more complex classifier for this novel biometric task. The distance-based approach calculates the Wasserstein distance between the extracted feature vectors. The machine learning approach learns the acquired active acoustic data through a combination of CNN and Transformer, classifying each sample according to which user it belongs to.

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3.1 Sweep signal generation

A sine wave whose frequency linearly changes from 20,001 Hz to 40,000 Hz over 0.2 seconds is generated as the sweep signal. Although human audible ranges vary with age, humans can generally hear frequencies roughly 10 Hz to 20,000 Hz. Therefore, the sound output from the speaker used is inaudible to humans. The sampling frequency of the generated sweep signal is 96 kHz.

3.2 Data acquisition

We use small piezoelectric elements (Measurement Specialties, model 11028362-00) as both the speaker (transmitter) and the microphone (receiver) to perform active acoustic sensing on the human body. While the sweep signal is being output, the response sound from the target body part is sampled at 96 kHz.

3.3 Preprocessing and feature extraction

Let $x[n]$ ($n = 0, \dots, L-1$) be the raw waveform. After pre-emphasis $x'[n] = x[n] - 0.97x[n-1]$ (boosting high frequencies), we segment the signal with 50%-overlapped Hamming windows of length $N = 1024$ and compute the DFT

$$X_i(k) = \sum_{n=0}^{N-1} x'[iH + n] w[n] e^{-j\frac{2\pi}{N}kn},$$

where i indexes frames and $H = N/2$. The frame power spectra $P_i(k) = |X_i(k)|^2$ are time-averaged to obtain a stable estimate $\bar{P}(k)$. We then isolate the bins corresponding to 20–40 kHz, normalize their energy, apply a log transform with $\epsilon = 10^{-8}$ to compress dynamic range, subtract the mean to remove channel bias, and stack the result into the feature vector $\hat{y} \in \mathbb{R}^M$.

3.4 Distance-based authentication

Let $\hat{y}^{(1)}, \hat{y}^{(2)} \in \mathbb{R}^M$ be the feature vectors

$$\hat{y}^{(j)} = [\hat{y}^{(j)}(1), \dots, \hat{y}^{(j)}(M)]^T, \quad j = 1, 2,$$

extracted from segments (M is the number of frequency bins). Their similarity is evaluated with the one-dimensional Wasserstein distance

$$W = \frac{1}{M} \sum_{k=1}^M \left| \sum_{m=1}^k \hat{y}^{(1)}(m) - \sum_{m=1}^k \hat{y}^{(2)}(m) \right|$$

which represents the optimal transport cost between the two cumulative spectra [10]. Authentication is granted when W is below a predefined threshold. We adopted the Wasserstein distance due to its robustness to small deviations in individual frequency bins and its ability to reflect global differences in spectral distributions. Although Dynamic Time Warping (DTW) is commonly used for aligning time sequences, our method relies on time-averaged spectra rather than raw time-series alignment. Therefore, we considered distribution-based comparison using the Wasserstein distance to be more appropriate.

3.5 CNN+Transformer classification

In this research, we aim to capture user-specific frequency characteristics and time-series features effectively from the time-frequency representation obtained by the short-time Fourier transform. Therefore, we employ a classification model that combines a 2D CNN

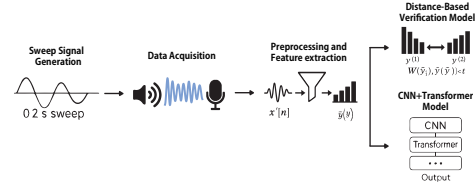


Figure 1: Overview of the two authentication pipelines.

(Convolutional Neural Network) with a Transformer. To capture user-specific spectral and temporal cues, we adopt a hybrid network that processes the STFT tensor $(B, 1, F, T)$. Three Conv2D – MaxPool2D blocks extract local features, which are flattened, regularized by dropout ($p = 0.3$), and projected to a 128-dimensional embedding. A Transformer encoder—with multi-head self-attention, feed-forward layers, LayerNorm, and dropout—models long-range dependencies, and a final linear layer plus softmax outputs class probabilities (B, C) . The network is trained with AdamW at a learning rate of 3×10^{-5} .

We employed a CNN+Transformer model to effectively capture localized time-frequency features through convolutional layers and temporal dynamics through self-attention. This configuration was chosen for its ability to learn complex, person-specific acoustic patterns.

4 Experiment

4.1 Data collection

Figure 2 depicts the fixture that secures the piezoelectric elements during recording. Measurements were taken at the finger, wrist, neck, nose, earlobe, ear bone, teeth, and sole—locations envisioned for future wearables. Indoors, a 20–40 kHz sweep was emitted for 30 seconds, and the response was sampled at 96 kHz. Seven subjects (A to G, all males, average age 21) recorded every site three times daily (morning after waking, midday after lunch, and evening before bed) over seven days. The acquired 30-second acoustic signal was subsequently divided into 2-second segments with no overlap, yielding 15 samples from each measurement. Consequently, for each subject and each site, the total number of samples was $(7 \text{ days}) \times (3 \text{ times}) \times (15 \text{ samples}) = 315 \text{ samples}$.

Using these samples, we conducted three types of evaluations. Evaluation 1 trains and tests within a single session (e.g., Day 1 morning). Evaluation 2 trains on all Day 1 sessions and tests on Days 2 to 7. Evaluation 3 trains on Days 1 to 6 and tests on Day 7. For distance-based approach, first 2-second interval (0–2s) of the 30-second data as the training data and the remainder as the test data. In the Machine learning approach pipeline, training data were augmented with a 100 ms sliding window, producing up to 280 windows per recording, while evaluation still used the fifteen disjoint segments.

4.2 Evaluation metrics

For the Distance-based approach, performance is measured by the equal error rate (EER), where the false-reject and false-accept rates



Figure 2: Devices used in the experiment.

Table 1: EER [%] by attachment site for Evaluation 1 using the distance-based approach.

Site	Subject						
	A	B	C	D	E	F	G
Finger	18.75	14.39	23.11	22.23	10.76	10.72	21.04
Wrist	15.23	13.27	18.45	15.39	15.09	15.48	17.67
Nose	21.71	1.73	17.98	13.66	9.58	14.52	10.55
Neck	15.40	6.90	17.82	22.26	6.80	12.72	12.54
Earlobe	14.82	3.74	7.03	8.90	14.12	16.44	10.50
Ear bone	20.52	6.33	17.34	12.42	6.75	8.90	14.83
Sole	18.60	11.69	20.01	5.14	11.65	3.70	13.59
Teeth	13.24	4.18	16.00	8.21	12.04	11.26	21.66

intersect. The machine learning approach is evaluated with the Macro-F1 score, the unweighted mean of per-class F1 values.

4.3 Results of distance-based approach

We present the results and discussions for Evaluations 1, 2, and 3 conducted using the distance-based authentication. In Evaluation 1, we selected data from a single day and a single time slot, e.g., the morning of Day 1, and divided it into training and test data. We performed this assessment for all 21 time slots (7 days $\times$ 3 times) and computed the mean and standard deviation of the EER across those time slots. The results are shown in Table 1. We can see that head-area sites, i.e., earlobe and ear bone, yielded relatively low EER values across all subjects. In particular, subject B exhibited remarkable stability at the nose (1.73%), earlobe (3.74%), and ear bone (6.33%). At extremities such as the finger or wrist, where sensor-to-skin contact tends to fluctuate, relatively high EER values of around 20% were observed, especially for subject C (23.11%) and subject D (22.23%). This suggests that authentication accuracy decreases in sites where the contact state of the sensor and skin is more prone to variation.

In Evaluation 2, we used the data from Day 1 (morning, midday, evening) as the training data and the data from Days 2 to 7 (morning, midday, evening) as the test data. Table 2 shows the EER values by site and subject. From these results, the EER varies considerably (approximately 30% to 60%) depending on the site and subject. In particular, the finger, sole, and nose yielded high EER values, indicating poor results. Some subjects showed relatively

Table 2: EER [%] by attachment site for Evaluation 2 using the distance-based approach.

Site	Subject						
	A	B	C	D	E	F	G
Finger	44.64	52.88	47.81	51.82	31.08	33.56	55.88
Wrist	32.74	50.41	41.98	46.15	48.01	44.26	47.74
Nose	49.80	15.30	51.57	51.63	39.02	36.32	35.75
Neck	48.41	39.99	50.99	60.24	35.68	32.99	42.14
Earlobe	51.29	23.26	58.43	50.15	49.60	48.07	42.55
Ear bone	48.19	21.71	54.20	52.75	50.24	21.94	47.69
Sole	47.07	52.94	52.48	44.00	39.19	48.40	40.08
Teeth	37.51	48.98	40.26	50.45	52.31	33.68	34.41

Table 3: EER [%] by attachment site for Evaluation 3 using the distance-based approach.

Site	Subject						
	A	B	C	D	E	F	G
Finger	40.03	66.09	52.64	56.99	31.72	39.08	46.36
Wrist	33.73	65.19	34.21	41.06	47.48	41.97	49.82
Sole	44.46	54.51	51.11	45.02	40.54	39.83	37.93
Nose	45.79	14.77	59.37	48.41	40.56	37.12	35.79
Neck	48.37	40.23	49.70	51.79	36.30	32.78	43.57
Earlobe	51.56	24.52	55.99	46.11	41.91	54.96	42.47
Ear bone	45.63	19.82	49.33	55.86	53.96	26.65	62.28
Teeth	25.60	47.63	37.78	43.41	54.30	32.05	45.43

lower EERs for teeth (37.51% to 52.31%) and wrist (32.74% to 50.41%), but Subject D displayed an EER of around 50% across all sites, revealing considerable individual differences. Moreover, those sites with high EER (finger, sole, nose) are places where the position of the piezoelectric element can easily shift at each measurement, leading to fluctuations in the data and thereby higher EER values.

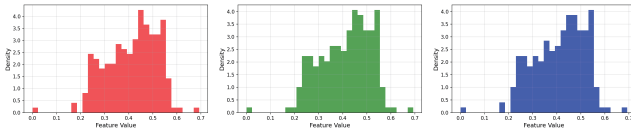
In Evaluation 3, we used the data from Days 1 to 6 (morning, midday, evening) as the training data and the data from Day 7 (morning, midday, evening) as the test data. Table 3 presents the EER for each site and subject. As in Evaluation 2, the EER values remain generally high. In particular, the finger and wrist exceed 60%. On the other hand, for the teeth site, Subject A recorded a relatively low EER of 25.60%, indicating that stability may differ by site. From these findings, it appears that even increasing the amount of training data over multiple days was insufficient to fully accommodate day-to-day variability in this experiment.

4.4 Results of Machine learning approach

Table 4 summarizes the CNN+Transformer results. In Evaluation 1, training and test data share the same session, so attachment and environment are constant; every site reaches a Macro-F1 of 100%. By contrast, Evaluations 2 and 3 use data from different days, and daily shifts in sensor position or body state markedly lower accuracy. In Evaluation 2, the best site sole scores only 24%, while wrist and earlobe fall to 18%. These sites likely suffer from contact instability. Adding six days of training data (Evaluation 3) partly mitigates the drop: nose rises to 51.3%, suggesting bone-conducted signals there

Table 4: Macro-F1 scores [%] for Evaluations 1 to 3 at each attachment site.

Site	Evaluation 1	Evaluation 2	Evaluation 3
Finger	100.0	19.0	46.2
Wrist	100.0	18.0	47.6
Nose	100.0	22.0	51.3
Neck	100.0	20.0	45.2
Earlobe	100.0	18.0	39.3
Sole	100.0	24.0	39.1
Ear bone	100.0	22.0	40.6
Teeth	100.0	20.0	28.6

**Figure 3: Feature distributions at three time segments within the same session (Day 1 morning): 0–2s (left), 14–16s (center), 28–30s (right).**

remain relatively consistent, whereas teeth stay lowest at 28.6%, perhaps due to intra-oral changes such as saliva or mouth opening. Overall, the model excels within sessions but falters across days because it cannot fully absorb temporal variation in anatomy and attachment. Future work should stabilize hardware, employ online adaptation, or augment data to offset placement shifts.

5 Discussion

Our multi-day experiments show that even slight shifts in sensor reattachment cause large variations in the recorded acoustic response, resulting in substantial drops in authentication accuracy. Prior on-body acoustic studies limited to same-day data achieved EERs below 10% [4, 5], whereas our cross-day results are markedly worse, underscoring the impact of temporal variability. We also observed intra-session distribution shifts (0–2s vs. 14–16s vs. 28–30s; see Fig 3), indicating sensitivity to contact pressure, adhesion, ambient noise, and subtle motion. By contrast, air-coupled approaches like SonicID [9] maintain stable cross-day performance, likely due to fixed device alignment.

To achieve long-term stability, improvements are required on both algorithmic and hardware fronts. Algorithmically, placement-aware modeling, domain adaptation, and self-correction mechanisms can mitigate attachment variability. Hardware solutions include fixtures or guides for repeatable placement and integrated pressure sensors for real-time coupling feedback. Finally, our participant pool (seven young males) is limited; future studies should recruit a more diverse cohort (varying age, gender, BMI, skin properties) to assess inter-subject differences in acoustic response.

In summary, active acoustic sensing shows promise for wearable authentication but demands robust algorithms, refined hardware designs, and broader validation to overcome attachment-induced instability.

6 Conclusion

We used active acoustic sensing to evaluate personal authentication performance at eight body sites. The experimental results show that performance at all sites deteriorates significantly over one or more days due to the effect of slight deviations in the contact position of the sensor. Future work will focus on redesigning the fixture for consistent skin contact, augmenting training with placement-robust data, and enlarging the participant pool to validate generalizability once hardware consistency is achieved. These improvements will be crucial to making on-body acoustic biometric authentication feasible in real-world wearable applications.

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